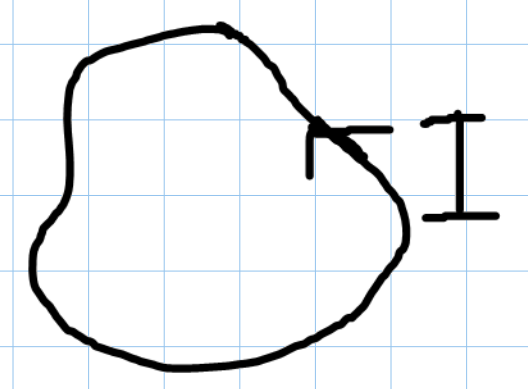
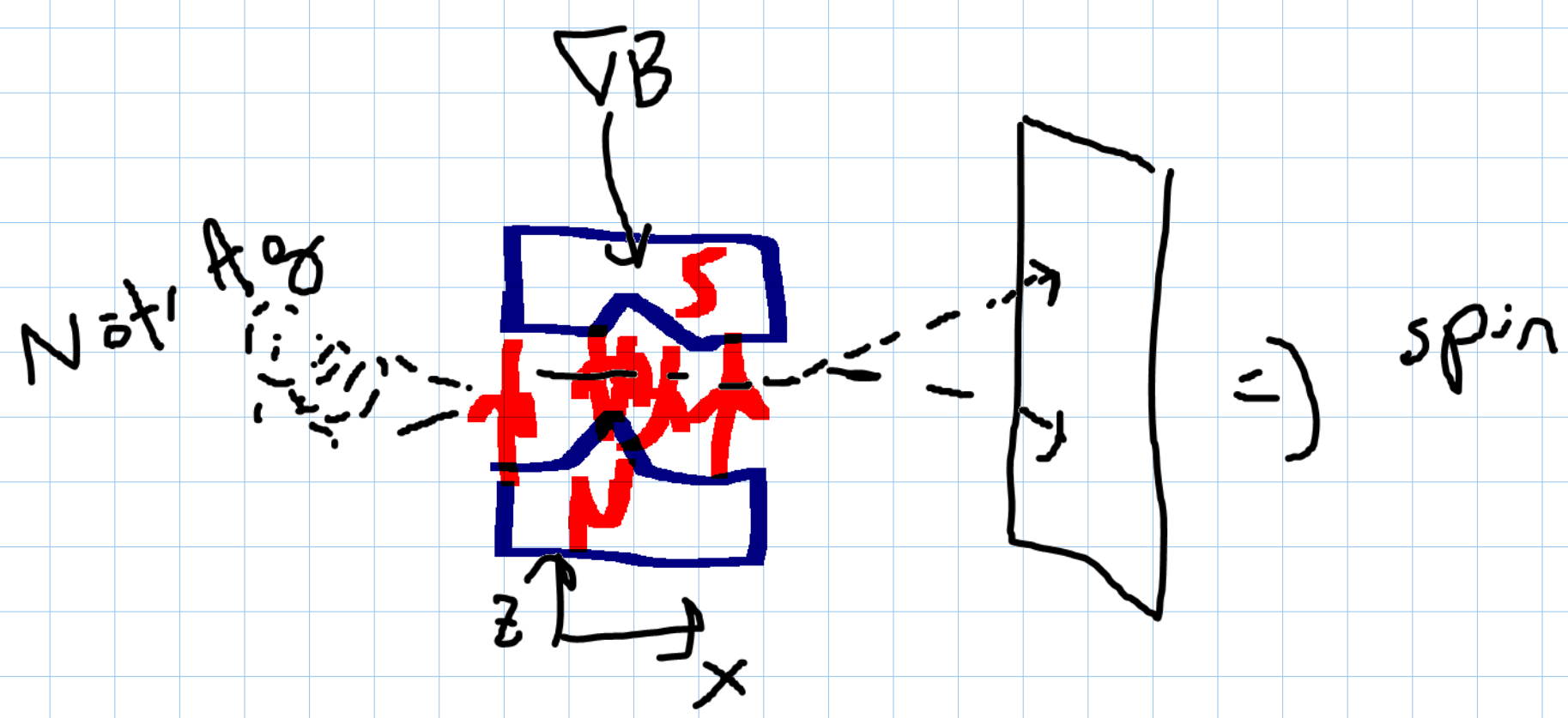


$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Stern Gerlach Deneyi (1921)



$$L = -\mu B$$

$$I \cdot A = \frac{Q}{T} A \Rightarrow \frac{Q}{2\pi r} \cdot \pi r^2 = \frac{Q r^2}{2} \Rightarrow \frac{Q r^2}{2} \cdot \frac{1}{r} = \frac{Q r}{2} = \mu_B$$

$$\frac{Q g S}{2 m} = \mu_B \quad S = \frac{\hbar}{2} \Rightarrow -\frac{e \hbar}{4 m_e}$$

$$\frac{Q g S}{2 m} = \mu_B$$

$$n \quad l \quad m \quad 2 \quad l=1, 0 \quad m=0 \Rightarrow l=1 \Rightarrow -1, 0, 1$$

$$\left(S = \frac{1}{2} \quad m_s = -\frac{1}{2}, +\frac{1}{2} \right)$$

$$S = S_x \quad S_y \quad S_z \Rightarrow S_x = \frac{\hbar}{2} \sigma_x \quad S_y = \frac{\hbar}{2} \sigma_y \quad S_z = \frac{\hbar}{2} \sigma_z$$

Soru $S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ S_z nin $\frac{\hbar}{2}$ ne vektörüdür?

$$S_z |\psi\rangle = \frac{\hbar}{2} |\psi\rangle \Rightarrow \psi = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \Rightarrow \alpha = \alpha \quad -\beta = \beta \Rightarrow 2\beta = 0 \quad \beta = 0 \Rightarrow \begin{pmatrix} \alpha \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha \\ 0 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} \alpha \\ 0 \end{pmatrix} = |\psi\rangle \quad \langle \psi | \psi \rangle = 1 \Rightarrow \psi = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \psi^* = \alpha^* (1, 0) \Rightarrow \langle \psi | \psi \rangle = \alpha \alpha^* (1, 0 | 1, 0) = 1$$

$$\alpha^2 = 1 \Rightarrow \psi = \begin{pmatrix} \alpha \\ 0 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |\psi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \leftarrow \text{spin yukarı}$$

$$\alpha^2 = 1 \Rightarrow \alpha = -1, 1, e^{i\theta}, \dots \rightarrow 1 \text{ alalım kolay olsun}$$

Soru $S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ S_z nin $-\frac{\hbar}{2}$ ne vektörüdür?

$$S_z |\psi\rangle = -\frac{\hbar}{2} |\psi\rangle \Rightarrow \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = -\frac{\hbar}{2} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \Rightarrow \alpha = -\alpha \quad -\beta = -\beta \Rightarrow 2\alpha = 0 \quad \alpha = 0 \quad \beta = \beta \Rightarrow \psi = \begin{pmatrix} 0 \\ \beta \end{pmatrix} \Rightarrow \psi = \begin{pmatrix} 0 \\ \beta \end{pmatrix}$$

$$\psi = \begin{pmatrix} 0 \\ \beta \end{pmatrix} \Rightarrow \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |\psi\rangle \quad \beta = 1$$

$$\langle \psi | \psi \rangle = 1 \Rightarrow \psi^* = \langle \psi | = \beta^* (0, 1) \Rightarrow \langle \psi | \psi \rangle = \beta^* \beta (0, 1 | 0, 1) = 1 \Rightarrow \beta^2 = 1$$

$$\beta = -1, +1, e^{i\theta}, \dots \rightarrow 1 \text{ en kolayıdır}$$

$$\beta = +1 \Rightarrow |\psi\rangle = \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |\psi\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \leftarrow \text{spin aşağı}$$

$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ = spin yukarı $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ = spin aşağı

$|\uparrow\rangle$

$|\downarrow\rangle$

$X_{\uparrow}$

$|0\rangle$

$|1\rangle$

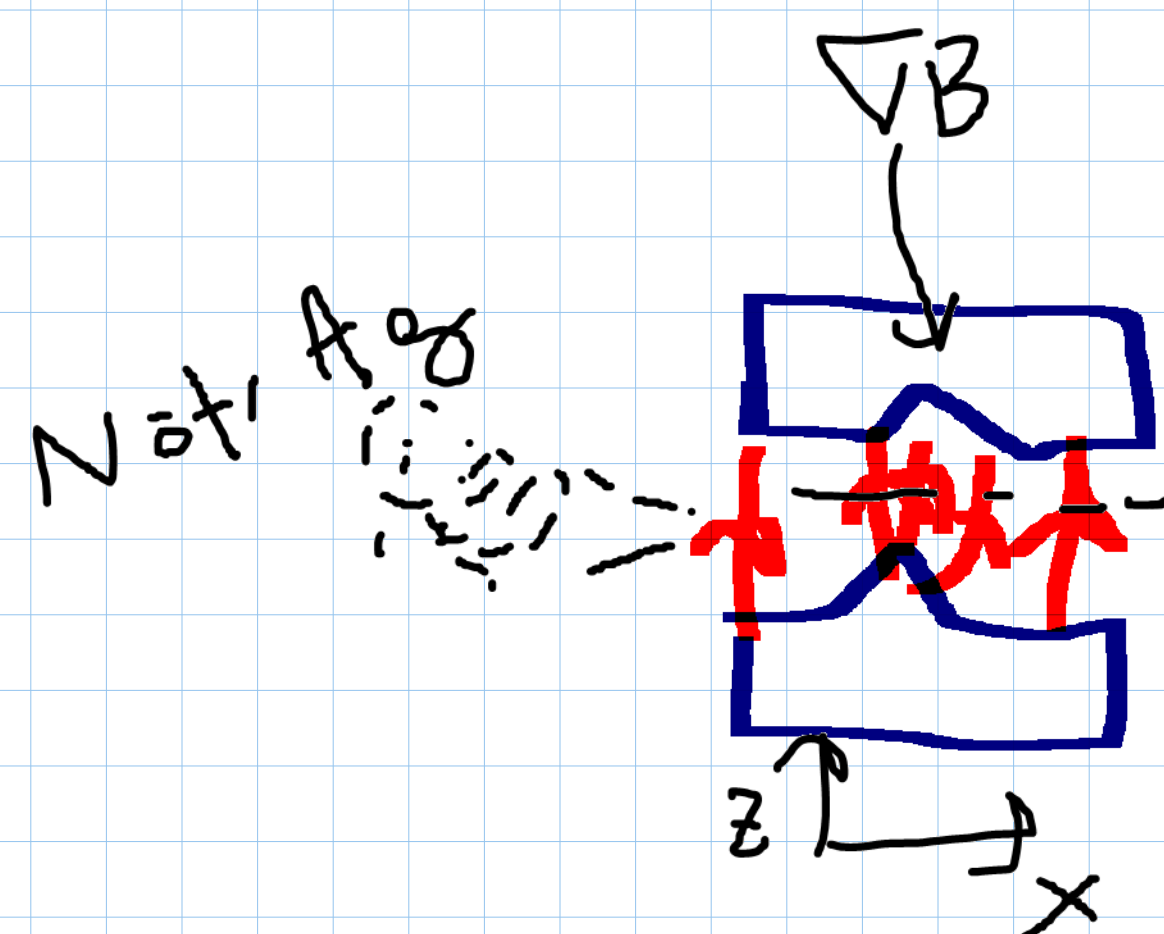
$X_{\downarrow}$

$|\frac{1}{2}, \frac{1}{2}\rangle$

$|\frac{1}{2}, -\frac{1}{2}\rangle$ ★

$s \nearrow m_s \nwarrow$

$s \nearrow m_s \nwarrow$

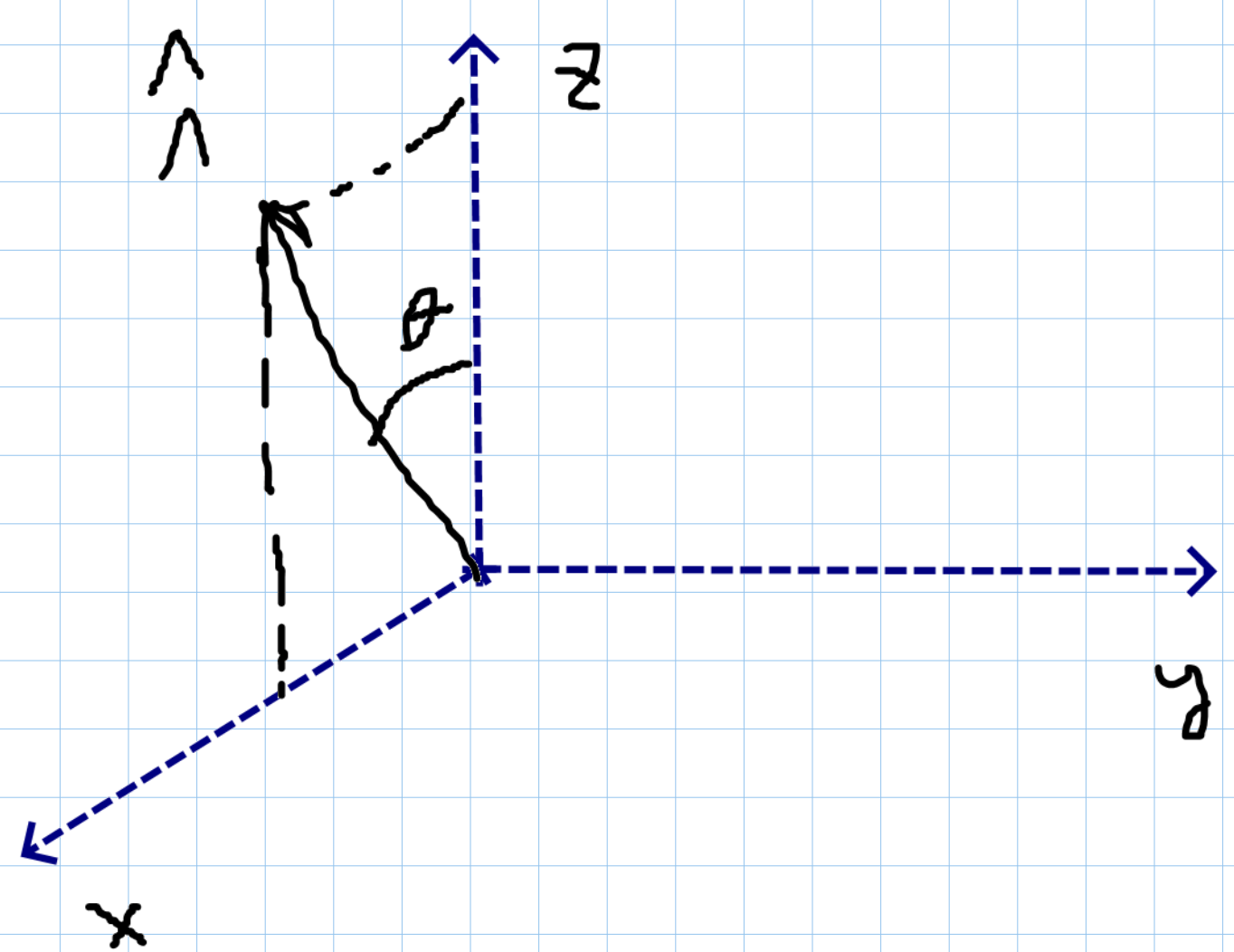


$$\Rightarrow \text{spin } A|\uparrow\rangle + B|\downarrow\rangle \Rightarrow \frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{1}{\sqrt{2}}|\downarrow\rangle = |+\rangle$$

$$\frac{1}{\sqrt{2}}|\uparrow\rangle - \frac{1}{\sqrt{2}}|\downarrow\rangle = |-\rangle$$

$$\Rightarrow \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = |+\rangle$$

$$\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle = |-\rangle$$



$$S_{\hat{n}} \Rightarrow S_{\hat{x}} + S_{\hat{z}}$$

$$S_n = \cos\theta S_z + \sin\theta S_x$$

$$\hat{n} = \cos\theta \hat{z} + \sin\theta \hat{x}$$

$$S_n = \cos\theta \frac{\hbar}{2} \sigma_z + \sin\theta \frac{\hbar}{2} \sigma_x \Rightarrow S_n = \frac{\hbar}{2} \left\{ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cos\theta + \sin\theta \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\} \Rightarrow \frac{\hbar}{2} \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$$

$$S_n = \frac{\hbar}{2} \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$$

$$S_z |\uparrow\rangle = \frac{\hbar}{2} |\uparrow\rangle \Rightarrow$$

$$\Rightarrow S_n |\uparrow\rangle = \frac{\hbar}{2} |\uparrow\rangle$$

$$|\uparrow\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$S_n |\uparrow\rangle \Rightarrow \frac{\hbar}{2} \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix} \cdot \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\Rightarrow \begin{aligned} \cos\theta \alpha + \sin\theta \beta &= \alpha \\ \sin\theta \alpha - \cos\theta \beta &= \beta \end{aligned} \Rightarrow \beta = \frac{\alpha(1 - \cos\theta)}{\sin\theta}$$

$$\cos\left(\frac{\theta}{2} + \frac{\theta}{2}\right) = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}$$

$$\sin\left(\frac{\theta}{2} + \frac{\theta}{2}\right) = \sin\frac{\theta}{2} \cos\frac{\theta}{2} + \cos\frac{\theta}{2} \sin\frac{\theta}{2}$$

$$\beta = \frac{\alpha(1 - \cos\theta)}{\sin\theta} = \frac{\alpha(1 - \cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2})}{\sin\frac{\theta}{2} \cos\frac{\theta}{2} + \cos\frac{\theta}{2} \sin\frac{\theta}{2}} \Rightarrow \frac{\cancel{\sin\frac{\theta}{2}} \cancel{\cos\frac{\theta}{2}} \sin\frac{\theta}{2}}{\cancel{\sin\frac{\theta}{2}} \cancel{\cos\frac{\theta}{2}} (\cancel{\cos\frac{\theta}{2}})} = \tan\frac{\theta}{2} \alpha$$

$$|\psi\rangle = |\uparrow\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \Rightarrow |\uparrow\rangle = \begin{pmatrix} \alpha \\ \tan\frac{\theta}{2} \alpha \end{pmatrix} \Rightarrow$$

$$|\uparrow\rangle = \alpha \begin{pmatrix} 1 \\ \tan\frac{\theta}{2} \end{pmatrix} \quad \langle \uparrow | = ? \Rightarrow \langle \uparrow | = \alpha^* \left(1 \quad \tan\frac{\theta}{2} \right)$$

$$\Rightarrow \langle \psi | \psi \rangle = 1 \Rightarrow \alpha \alpha^* \left(1, \tan\frac{\theta}{2} \right) \begin{pmatrix} 1 \\ \tan\frac{\theta}{2} \end{pmatrix} = |\alpha|^2 (1 + \tan^2 \frac{\theta}{2}) = 1 \Rightarrow |\alpha|^2 \left(\frac{1}{\cos^2 \frac{\theta}{2}} \right) = 1 \Rightarrow$$

$$|\alpha|^2 = \cos^2 \frac{\theta}{2} \Rightarrow \alpha = \cos\frac{\theta}{2}$$

$$|\psi\rangle = \alpha \begin{pmatrix} 1 \\ \tan \frac{\theta}{2} \end{pmatrix} \Rightarrow |\psi\rangle = \cos \frac{\theta}{2} \begin{pmatrix} 1 \\ \tan \frac{\theta}{2} \end{pmatrix} \Rightarrow |\psi\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{pmatrix} = |\uparrow\rangle$$

$$\rightarrow S_n |\uparrow\rangle = \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \cdot \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = -\frac{\hbar}{2} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\Rightarrow \cos \theta \alpha + \sin \theta \beta = -\alpha \quad \beta = -\alpha \frac{(1 + \cos \theta)}{\sin \theta}$$

$$\cos\left(\frac{\theta}{2} + \frac{\theta}{2}\right) = \cos \frac{\theta}{2} \cos \frac{\theta}{2} - \sin \frac{\theta}{2} \sin \frac{\theta}{2}$$

$$\sin\left(\frac{\theta}{2} + \frac{\theta}{2}\right) = \sin \frac{\theta}{2} \cos \frac{\theta}{2} + \cos \frac{\theta}{2} \sin \frac{\theta}{2}$$

$$\sin \theta \alpha - \cos \theta \beta = -\beta \Rightarrow \beta = -\alpha \frac{(1 + \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2})}{\sin \theta} = -\alpha \frac{2 \cos^2 \frac{\theta}{2}}{2 \cos \frac{\theta}{2} \sin \frac{\theta}{2}} = -\alpha \cot \frac{\theta}{2}$$

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha \\ -\alpha \cot \frac{\theta}{2} \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ -\cot \frac{\theta}{2} \end{pmatrix} = |\downarrow\rangle = |\psi\rangle$$

$$\langle \psi | \psi \rangle = 1 \quad \langle \psi | = \alpha^* (1, -\cot \frac{\theta}{2})$$

$$\langle \psi | \psi \rangle = 1 = \alpha \alpha^* \begin{pmatrix} 1, -\cot \frac{\theta}{2} \end{pmatrix} \begin{pmatrix} 1 \\ -\cot \frac{\theta}{2} \end{pmatrix} = (1 + \cot^2 \frac{\theta}{2}) = 1 \Rightarrow \alpha^2 \frac{1}{\sin^2 \frac{\theta}{2}} = 1 \Rightarrow \alpha = \sin \frac{\theta}{2}$$

$$\sin \frac{\theta}{2} \begin{pmatrix} 1 \\ -\cot \frac{\theta}{2} \end{pmatrix} = \begin{pmatrix} \sin \frac{\theta}{2} \\ -\cos \frac{\theta}{2} \end{pmatrix} = |\downarrow\rangle$$

hiz doğru

Soru $|x\rangle = \frac{1}{\sqrt{2}} (|\frac{1}{2}, \frac{1}{2}\rangle + |\frac{1}{2}, -\frac{1}{2}\rangle)$ bu vektörün S_x op'ü nedir?

$$S_x |x\rangle = \frac{\hbar}{2} |x\rangle = ? \quad \begin{aligned} S_+ &= S_x + iS_y \\ S_- &= S_x - iS_y \end{aligned} \Rightarrow 2S_x = S_+ + S_- \Rightarrow S_x = \frac{S_+ + S_-}{2}$$

$$S_+ |s, m_s\rangle = \hbar \sqrt{s(s+1) - m_s(m_s+1)} |s, m_s+1\rangle$$

$$S_- |s, m_s\rangle = \hbar \sqrt{s(s+1) - m_s(m_s-1)} |s, m_s-1\rangle$$

$$S_+ |\frac{1}{2}, \frac{1}{2}\rangle + S_+ |\frac{1}{2}, -\frac{1}{2}\rangle = \hbar \sqrt{\frac{1}{2}(\frac{1}{2}) - (-\frac{1}{2})(-\frac{1}{2}+1)} |\frac{1}{2}, -\frac{1}{2}+1\rangle = \sqrt{\frac{1}{4} + \frac{1}{4}} |\frac{1}{2}, \frac{1}{2}\rangle$$

$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ $-\frac{1}{2} + 1 = \frac{1}{2}$ ✓

$$S_+ |x\rangle = \frac{1}{\sqrt{2}} |x_{\uparrow}\rangle$$

$$S_- |s, m_s\rangle = S_- |x_{\uparrow}\rangle + S_- |x_{\downarrow}\rangle = \frac{1}{\sqrt{2}} |x_{\downarrow}\rangle$$

$$S_x |x\rangle = \frac{\hbar}{2} \left(\frac{1}{\sqrt{2}} |x_{\downarrow}\rangle + \frac{1}{\sqrt{2}} |x_{\uparrow}\rangle \right) = \frac{\hbar}{2} |x\rangle \Rightarrow S_x |x\rangle = \frac{\hbar}{2} |x\rangle$$

$$|x\rangle = \frac{1}{\sqrt{2}} |x_{\uparrow}\rangle + \frac{1}{\sqrt{2}} |x_{\downarrow}\rangle$$

Soln $|x\rangle = \frac{1}{\sqrt{2}} \left(\left| \frac{1}{2}, \frac{1}{2} \right\rangle + \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \right)$ bu vektörün S_x op'ün netir?

$$S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow |x\rangle = \frac{1}{\sqrt{2}} \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

$$\Rightarrow S_x |x\rangle = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) = \frac{\hbar}{2\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{\hbar}{2\sqrt{2}} \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) = \frac{\hbar}{2} |x\rangle$$

Soln $|x'\rangle = \frac{1}{\sqrt{2}} \left(|\uparrow\rangle - |\downarrow\rangle \right)$ $S_x |x'\rangle = ?$

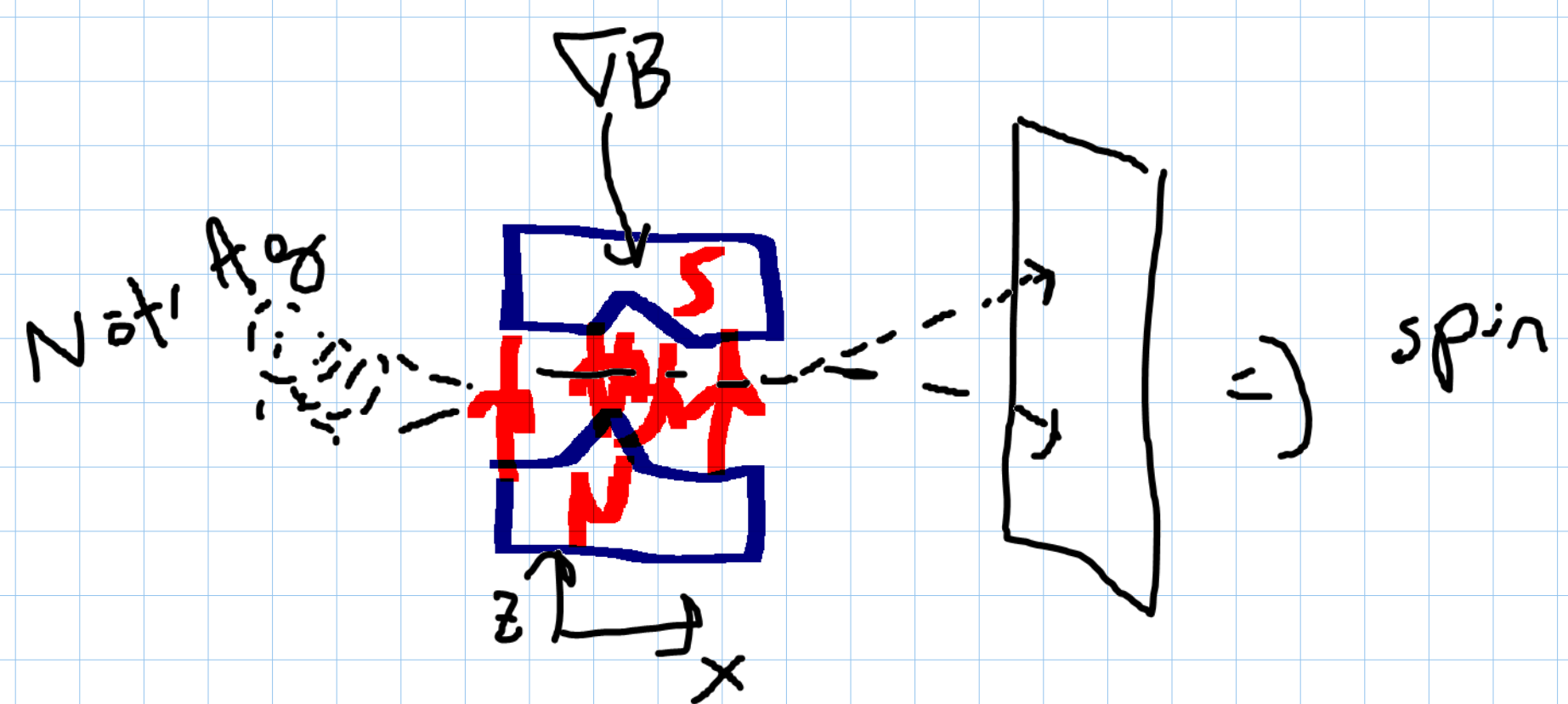
$$S_x = \frac{\hbar}{2} \sigma_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow \frac{1}{\sqrt{2}} \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\Rightarrow \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right) = \frac{\hbar}{2\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = S_x |x'\rangle \Rightarrow$$

$$|x'\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \Rightarrow S_x |x'\rangle = \frac{\hbar}{2} \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = - \begin{pmatrix} 1 \\ -1 \end{pmatrix} = - |x'\rangle$$

$$\Rightarrow S_x |x'\rangle = -\frac{\hbar}{2} |x'\rangle$$

$$\Rightarrow S_x |x'\rangle = -\frac{\hbar}{2} |x'\rangle$$



$$|+\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle + |\downarrow\rangle)$$

$$|-\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle - |\downarrow\rangle)$$

$$|\uparrow\rangle_x = \frac{1}{\sqrt{2}} (|\uparrow\rangle_z + |\downarrow\rangle_z)$$

$$|\downarrow\rangle_x = \frac{1}{\sqrt{2}} (|\downarrow\rangle_z - |\uparrow\rangle_z)$$

